

COMP 3270 – Introduction to Algorithms

Homework 8

Topics – Algorithm strategies, Hash tables

1. A rotation of an array is an operation in which the elements of the array are shifted left or right by some number of positions, where elements that “fall off” one side of the array are added back to the other side, in the same order. For example, if the array containing the elements 1-10 (in order) were shifted right by 4 positions (or left 6), it would contain:

7	8	9	10	1	2	3	4	5	6
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The Inverse Rotation problem gives you an array (of length n) that is the result of rotating a sorted array k positions to the right, and the solution to the problem is to identify k . (This solution is also the number of positions to rotate the given array left to make it sorted again.) If the input to the Inverse Rotation problem is already sorted, the correct solution is 0; however, if the input array is not a rotation of a sorted array (such as the array [3 2 1]), any output is acceptable.

Give pseudocode for an $O(n)$ brute-force algorithm to solve the Inverse Rotation problem. You may assume that the input array contains no duplicate values.

2. Consider inserting the keys 10, 22, 31, 4, 15, 28, 17, 88, 59 into a hash table of length $m = 11$ using open addressing with the auxiliary hash function $h'(k) = k$. Illustrate the result of inserting these keys using linear probing, using quadratic probing with $c_1 = 1$ and $c_2 = 3$. Use the hash function of $h(k) = k \bmod m$, for quadratic probing, $h(k) = (h(k) + c_1 i + c_2 i^2)$.
3. Based on the given scenario, which algorithmic strategy (Divide and conquer, scatter and gather, brute force, data structures etc.,) would be the most suitable choice, and why?
 - a. You are developing a password security system and need to test how secure a user's password is. The system must check whether the password can be cracked by trying all possible combinations of characters.
 - b. A satellite imaging company needs to process a large dataset of high-resolution images to detect deforestation patterns. Each image is divided into sections, and the algorithm must efficiently identify regions where trees have been removed.
 - c. You work for a large e-commerce company that needs to analyze millions of customer reviews to determine the most common complaints about a product. The dataset is too large to process on a single machine.
 - d. You are designing a web browser that supports the "Back" and "Forward" navigation buttons to allow users to revisit previously visited web pages.

- e. A call center receives customer service requests, and each request must be processed in the order it was received. Some requests take longer than others to resolve.
- f. An emergency room receives patients with different levels of severity, ranging from minor injuries to life-threatening conditions. Patients must be treated based on urgency rather than arrival time.
- g. You are building an autocomplete feature for a search engine that suggests the most common search queries based on user input. The system must retrieve relevant suggestions in real-time.
- h. An airline booking system needs to assign seats based on passenger status—first-class passengers get priority over economy-class passengers.
- i. A music streaming app needs to provide instant song recommendations based on a user's recent searches. The system should store frequently searched songs for quick retrieval.
- j. A hospital wants to match patients with the most compatible organ donors based on multiple criteria, including blood type, genetic markers, and medical history. A naïve approach would be to compare every patient with every donor. What will be your approach?